ANSWER KEY

25900139 **B**

Reg.No : .....
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
NA SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCMAT100 - APPLICABLE MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 1.25 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple type Questions

Answer any **25** questions

Each question carries **2** marks

1. B: $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$
2. A:Both PQ and QP are defined.
3. A: $\frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x))$
4. D:Skew-symmetric matrix
5. B: $2 - 2x$
6. C: 3
7. A: 0
8. For any answer, give full marks
9. B: $\frac{m(m-1)(m-2)}{6}$
10. B: 8^8
11. B: 0

12. B:3×2

13. D:$1, 7$

14. C:$x + 1$

15. C:$-x \sin x + \cos x$

16. A:3 Elements

17. A:$2x + 1$

18. C:0

19. C:$\sec^2 x$

20. A:n + 1

21. B:$\begin{bmatrix} 9 & 3 & 6 \\ 18 & -3 & 24 \end{bmatrix}$

22. A:$(3x - 2)(x + 4)$

23. D:$\begin{bmatrix} 14 & 14 \\ 13 & 12 \end{bmatrix}$

24. D:2520

25. B:$\begin{bmatrix} 0 & 1 & 0 \\ -3 & 6 & -2 \\ 7 & -3 & -3 \end{bmatrix}$

26. B:24

27. B:$8x \sec^2(4x^2)$

28. A:45

29. D:$\frac{(n-1)!}{n+1}$

30. B:$\frac{^{12}C_3 \times ^9C_3 \times ^6C_3 \times ^3C_3}{4!}$

[2x25 = 50]